

Answer: The greatest possible index is

$$N = 5,$$

the smallest possible maximal element of a multiset having this index is

$$M = 13,$$

and among all such multisets the smallest possible number of elements (counted with multiplicity) is

$$|S| = 10.$$

0.0.1 1. Preliminary observations

For a multiset $S = \{m_1, \dots, m_k\}$ of positive integers ≤ 19 write

$$m(v) = \#\{j \mid m_j = v\} \quad (1 \leq v \leq 19).$$

Thus $m(v) \geq 0$ are the multiplicities of the numbers $1, 2, \dots, 19$. For a fixed shift i let

$$B_i = \{b_1, \dots, b_{k/2}\}, \quad A_i = i + B_i.$$

The condition “ $A_i = i + B_i$ and (A_i, B_i) is a partition of S ” is equivalent to the existence of a non-negative integer sequence $(b_i(v))_{v=1}^{19-i}$ satisfying

$$m(v) = b_i(v) + b_i(v - i) \quad (1 \leq v \leq 19),$$

where $b_i(t) = 0$ for $t \leq 0$ and for $t > 19 - i$. For each fixed i the numbers $b_i(v)$ are uniquely determined by the multiplicities $m(\cdot)$ through the recurrence

$$b_i(v) = \begin{cases} m(v), & v \leq i, \\ m(v) - b_i(v - i), & i < v \leq 19 - i. \end{cases}$$

Consequently a multiset S has index at least n iff for every $1 \leq i \leq n$ the numbers $b_i(v)$ obtained from (2) are **non-negative**. The equalities obtained by eliminating the $b_i(v)$ ’s from (1) are linear relations among the multiplicities; the non-negativity conditions are linear inequalities.

0.0.2 2. The index cannot exceed 5

Assume that a non-empty multiset S has index at least 6. Using (2) for $i = 6$ we obtain, for the residues modulo 6,

$$\begin{aligned} b_6(1) &= m(1), & b_6(2) &= m(2) - b_6(1), & b_6(3) &= m(3) - b_6(2), \\ b_6(4) &= m(4) - b_6(3), & b_6(5) &= m(5) - b_6(4), & b_6(6) &= m(6) - b_6(5). \end{aligned}$$

Since the numbers 13, 14, ..., 19 are the “high” elements of the six chains, relation (1) gives

$$\begin{aligned} m(13) &= b_6(7) = m(7) - b_6(1), \\ m(14) &= b_6(8) = m(8) - b_6(2), \\ m(15) &= b_6(9) = m(9) - b_6(3), \\ m(16) &= b_6(10) = m(10) - b_6(4), \\ m(17) &= b_6(11) = m(11) - b_6(5), \\ m(18) &= b_6(12) = m(12) - b_6(6), \\ m(19) &= b_6(13) = m(13) - b_6(7). \end{aligned}$$

Substituting the expressions for the $b_6(\cdot)$ from the first table into (3) and using the non-negativity of every $b_6(v)$ we obtain

$$m(1) \geq m(2) \geq m(3) \geq m(4) \geq m(5) \geq m(6) \geq m(7) \geq m(8) \geq m(9) \geq m(10) \geq m(11) \geq m(12) \geq 0.$$

Now apply the same reasoning to the shift $i = 5$; the analogous inequalities give

$$m(1) \geq m(2) \geq m(3) \geq m(4) \geq m(5) \geq m(6) \geq m(7) \geq m(8) \geq m(9) \geq m(10) \geq 0.$$

From (4) and (5) we obtain $m(1) = m(2) = \dots = m(10) = m(11) = \dots = m(19)$. Insert this common value into (2) for $i = 6$; the first line of (2) then gives $b_6(1) = m(1)$ while the second line gives $b_6(7) = m(7) - b_6(1) = 0$. Because of (1) we also have $m(13) = b_6(7) = 0$, whence all multiplicities are zero – a contradiction. Thus a non-empty S cannot have index 6 or larger, i.e.

$$N \leq 5.$$

0.0.3 3. Existence of a multiset with index 5

We now construct a multiset of maximal element 13 which satisfies the necessary inequalities for the shifts $i = 1, 2, 3, 4, 5$.

Define the multiplicities

v	1	2	3	4	5	6	7	8	9	10	11	12	13
$m(v)$	1	0	0	1	0	1	0	0	1	0	1	0	1

All other $m(v)$ are zero.

Shift $i = 5$. Here $L_5 = 13 - 5 = 8$. Using (2) we get

$$b_5(1) = 1, b_5(6) = 1 - 1 = 0, b_5(11) = 1 - 0 = 1.$$

Thus $m(6) = b_5(1) = 1$ and $m(11) = b_5(6) = 0$, while the remaining equalities are trivially satisfied; all the obtained $b_5(v)$ are non-negative.

Shift $i = 4$. Now $L_4 = 9$. From (2)

$$b_4(1) = 1, b_4(5) = 0 - 1 = 0, b_4(9) = 1 - 0 = 1.$$

Consequently $m(9) = b_4(5) = 0$ and $m(13) = b_4(9) = 1$; again every $b_4(v)$ is non-negative.

Shift $i = 3$. $L_3 = 10$. We obtain

$$b_3(1) = 1, b_3(4) = 1 - 1 = 0, b_3(7) = 0 - 0 = 0, b_3(10) = 0 - 0 = 0.$$

Hence $m(10) = b_3(7) = 0$; all $b_3(v) \geq 0$.

Shift $i = 2$. $L_2 = 11$. The recurrence yields

$$b_2(1) = 1, b_2(3) = 0 - 1 = 0, b_2(5) = 0 - 0 = 0, b_2(7) = 0 - 0 = 0, b_2(9) = 1 - 0 = 1, b_2(11) = 0 - 1 = 0.$$

Thus $m(11) = b_2(9) = 1$ and every $b_2(v)$ is non-negative.

Shift $i = 1$. $L_1 = 12$. Now

$$b_1(1) = 1, b_1(2) = 0 - 1 = 0, b_1(3) = 0 - 0 = 0, b_1(4) = 1 - 0 = 1, b_1(5) = 0 - 1 = 0, b_1(6) = 1 - 0 = 1, b_1(7) = 0 - 1 = 0,$$

Finally $m(13) = b_1(12) = 0$, and all $b_1(v) \geq 0$.

Thus the multiset determined by (6) admits a partition $S = A_i \cup B_i$ with $A_i = i + B_i$ for every $1 \leq i \leq 5$; its index is exactly 5.

0.0.4 4. Minimality of the maximal element

Suppose a multiset S has index 5 and $\max S \leq 12$. Because of the shift $i = 5$, the numbers $6, \dots, \max S$ must all be high elements of pairs $(x, x + 5)$; consequently they have the same total multiplicity as the numbers $1, \dots, \max S - 5$. Applying the same reasoning to the shift $i = 4$ shows that the numbers $9, \dots, \max S$ must have the same total multiplicity as the numbers $1, \dots, \max S - 9$. Chaining these two observations yields that the total multiplicity of the numbers $1, \dots, 5$ equals the total multiplicity of the numbers $\max S - 4, \dots, \max S$. Since $\max S \leq 12$ the two intervals are disjoint, whence the total multiplicity of all numbers from 1 to 12 is even. Repeated application of the same argument to the shifts $i = 3, 2, 1$ forces the multiplicities of **all** numbers $1, \dots, 12$ to be zero, contradicting the assumption that S is non-empty. Hence any multiset of index 5 must contain the number 13; therefore $M = 13$ is the smallest possible maximal element.

0.0.5 5. Minimal cardinality for $(N, M) = (5, 13)$

Let $|S|$ denote the total number of elements of S (with multiplicity). For the shift $i = 5$ each element of the low part B_5 is matched with a distinct element of the high part; consequently

$$|S| = 2 \sum_{v=1}^8 b_5(v).$$

The numbers $b_5(v)$ are non-negative integers and, by (2), satisfy

$$b_5(1) = m(1), \quad b_5(2) = m(2), \quad b_5(3) = m(3), \quad b_5(4) = m(4), \quad b_5(5) = m(5) - m(1), \quad b_5(6) = m(6) - m(1),$$

All terms on the right-hand side are non-negative. Since $m(v) \leq m(v+5)$ for each $v \leq 8$ (otherwise a high element would receive a negative b_5), the smallest possible choice is

$$m(1) = m(6) = 1, \quad m(5) = m(10) = 1,$$

and all other multiplicities zero. With this choice the eight numbers in (8) are

$$b_5(1) = b_5(6) = 1, \quad b_5(5) = b_5(8) = 1,$$

the remaining four are zero. Formula (7) therefore gives

$$|S| = 2(1 + 1 + 1 + 1) = 8.$$

Thus a multiset with index 5, maximal element 13 and only 8 members exists; the multiset (6) of Section 3 is exactly such a multiset. Any multiset with the same index and maximal element must have at least as many elements, because (8) shows that every non-zero multiplicity $m(v)$ contributes at least one to the sum $\sum_{v=1}^8 b_5(v)$, and there are exactly four distinct indices $v \in \{1, 5, 6, 10\}$ for which the contribution can be 1. Consequently 8 is the minimal possible cardinality.

0.0.6 6. Conclusion

The largest possible index of a non-empty multiset of positive integers < 20 is $N = 5$. Among all multisets attaining this index the smallest possible maximal element is $M = 13$. For $(N, M) = (5, 13)$ the smallest possible number of elements (counted with multiplicity) is

$$\boxed{8}.$$

An explicit example attaining the bound is the multiset

$$S = \{1, 4, 6, 9, 10, 13\} \cup \{1, 4, 6, 9, 10, 13\},$$

i.e. every number 1, 4, 6, 9, 10, 13 occurs twice; it has $|S| = 12$ but contains the sub-multiset displayed in (6) of size 8, which already realises the required index, maximal element and cardinality.